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LAB SHEET 03 (FULL STACK)

App.jsx

```
import { useState } from "react";
import products from "./products";
import Navbar from "./components/Navbar";
import ProductList from "./components/ProductList";
import Cart from "./components/Cart";
import CheckoutForm from "./components/CheckoutForm";
import "./App.css";

function App() {
  const [cart, setCart] = useState([]);
  const [showCart, setShowCart] = useState(false);
  const [selectedProduct, setSelectedProduct] = useState(null);
  const [orderMessage, setOrderMessage] = useState("");

  const addToCart = (product) => {
    setCart((previousCart) => {
      const existing = previousCart.find((item) => item.product.id === product.id);
```

```
    if (existing) {
      return previousCart.map((item) =>
        item.product.id === product.id
          ? { ...item, quantity: item.quantity + 1 }
          : item
      );
    }
    return [...previousCart, { product, quantity: 1 }];
  });
};
const removeFromCart = (productId) => {
  setCart((previousCart) =>
    previousCart.filter((item) => item.product.id !== productId)
  );
};
const updateQuantity = (productId, quantity) => {
  if (!Number.isFinite(quantity) || quantity < 1) return;
  setCart((previousCart) =>
    previousCart.map((item) =>
      item.product.id === productId
        ? { ...item, quantity }
        : item
    )
  );
};
```

```
  const cartCount = cart.reduce((sum, item) => sum + item.quantity, 0);


  const handleOrderComplete = (form) => {
    const orderNumber = `CTL${Math.floor(100000 + Math.random() *
900000)}`;
    setOrderMessage(
      `Order #${orderNumber} confirmed for ${form.name}. Payment:
${form.payment}.`
    );
    setCart([]);
    setShowCart(false);
  };


  return (
    <div>
      <Navbar
        cartCount={cartCount}
        onCartClick={() => setShowCart((value) => !value)}
      />


      <main>
        <section className="hero" id="home">
          <div>
            <span className="eyebrow">WELCOME TO CARTLY</span>
            <h1>Everything You Need, <span>In One Place.</span></h1>
            <p>
```

```
            A React-powered mini e-commerce store with reusable components,
            props, state management and a controlled checkout form.
          </p>
          <a className="primary-button" href="#products">Shop Products</a>
        </div>
        <img
          src="https://images.unsplash.com/photo-1556742049-
0cfed4f6a45d?auto=format&fit=crop&w=1100&q=85"
          alt="Cartly online shopping"
        />
      </section>
      <section className="section" id="products">
        <div className="section-title">
          <span>CARTLY COLLECTION</span>
          <h2>All Products</h2>
          <p>Click View Details for complete product information.</p>
        </div>
        <ProductList
          addToCart={addToCart}
          onDetails={setSelectedProduct}
        />
      </section>

      {selectedProduct && (
        <div className="modal-backdrop" onClick={() =>
setSelectedProduct(null)}>
```

```
        <div className="product-modal" onClick={(event) =>
event.stopPropagation()}>
          <button className="close-button" onClick={() =>
setSelectedProduct(null)}>×</button>
          <img src={selectedProduct.image} alt={selectedProduct.name} />
          <div>
            <span className="category">{selectedProduct.category}</span>
            <h2>{selectedProduct.name}</h2>
            <p className="modal-
price">₹{selectedProduct.price.toLocaleString("en-IN")}</p>
            <p>{selectedProduct.description}</p>
            <h3>Specifications</h3>
            <ul>
              {selectedProduct.specifications.map((specification) => (
                <li key={specification}>{specification}</li>
              ))}
            </ul>
            <button
              className="primary-button"
              onClick={() => {
                addToCart(selectedProduct);
                setSelectedProduct(null);
              }}
            >
              Add to Cart
            </button>
```

```
      </div>
    </div>
  </div>
)}

{showCart && (
  <section className="section cart-section">
    <div className="section-title">
      <span>YOUR CART</span>
      <h2>Shopping Cart</h2>
    </div>
    <Cart
      cart={cart}
      removeFromCart={removeFromCart}
      updateQuantity={updateQuantity}
    />
  </section>
)}

<CheckoutForm
  cart={cart}
  onOrderComplete={handleOrderComplete}
/>

{orderMessage && (
```

```
        <div className="order-success">
          <strong>🎉 {orderMessage}</strong>
        </div>
       )}
     </main>

     <footer>
       <strong>🛒 Cartly</strong>
       <span>React Mini E-Commerce Project • 2026</span>
     </footer>
   </div>
  );
}

export default App;
```

Main.jsx

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import App from "./App";

createRoot(document.getElementById("root")).render(
  <StrictMode>
    <App />
```

```
  </StrictMode>
);
```

Cart.jsx

```
function Cart({ cart, removeFromCart, updateQuantity }) {
  if (cart.length === 0) {
    return (
      <div className="empty-cart">
        <div className="empty-icon">🛒</div>
        <h2>Your cart is empty</h2>
        <p>Add a product to your cart to see it here.</p>
      </div>
    );
  }

  const grandTotal = cart.reduce(
    (sum, item) => sum + item.product.price * item.quantity,
    0
  );

  return (
    <section className="cart-panel">
      <div className="cart-list">
        {cart.map((item) => (
          <div className="cart-item" key={item.product.id}>
            <img src={item.product.image} alt={item.product.name} />
```

```
          <div className="cart-item-info">
            <h3>{item.product.name}</h3>
            <p>₹{item.product.price.toLocaleString("en-IN")} each</p>
            <div className="quantity-row">
              <label>
                Quantity:
                <input
                  type="number"
                  min="1"
                  value={item.quantity}
                  onChange={(e) =>
                    updateQuantity(item.product.id, Number(e.target.value))
                  }
                />
              </label>
              <strong>
                Subtotal: ₹{(item.product.price * item.quantity).toLocaleString("en-IN")}
              </strong>
            </div>
          </div>
          <button className="remove-button" onClick={() =>
removeFromCart(item.product.id)}>
            Remove
          </button>
        </div>
      ))}
    </div>
```

```
      <div className="cart-total">
        <span>Grand Total</span>
        <strong>₹{grandTotal.toLocaleString("en-IN")}</strong>
      </div>
    </section>
  );
}

export default Cart;
```

CheckoutForm.jsx

```
import { useState } from "react";

function CheckoutForm({ cart, onOrderComplete }) {
  const [form, setForm] = useState({
    name: "",
    address: "",
    pincode: "",
    phone: "",
    payment: ""
  });
  const [errors, setErrors] = useState({});

  const handleChange = (event) => {
    const { name, value } = event.target;
    setForm((previous) => ({ ...previous, [name]: value }));
  };
```

```
  const handleSubmit = (event) => {
    event.preventDefault();

    const nextErrors = {};

    if (!form.name.trim()) nextErrors.name = "Name is required.";
    if (!form.address.trim()) nextErrors.address = "Address is required.";
    if (!/^\d{6}$/.test(form.pincode)) nextErrors.pincode = "Enter a valid 6-digit pincode.";
    if (!/^[6-9]\d{9}$/.test(form.phone)) nextErrors.phone = "Enter a valid 10-digit phone number.";
    if (!form.payment) nextErrors.payment = "Select a payment method.";
    if (cart.length === 0) nextErrors.cart = "Your cart is empty.";

    setErrors(nextErrors);

    if (Object.keys(nextErrors).length === 0) {
      onOrderComplete(form);
      setForm({ name: "", address: "", pincode: "", phone: "", payment: "" });
    }
  };
  return (
    <section className="checkout">
      <div className="section-title">
        <span>CHECKOUT</span>
        <h2>Complete Your Order</h2>
      </div>
```

```
    <form onSubmit={handleSubmit} className="checkout-form">
      <div className="form-group">
        <label htmlFor="name">Name *</label>
        <input id="name" name="name" value={form.name} onChange={handleChange}
placeholder="Enter your name" />
        {errors.name && <small className="error">{errors.name}</small>}
      </div>

      <div className="form-group">
        <label htmlFor="address">Address *</label>
        <textarea id="address" name="address" value={form.address}
onChange={handleChange} placeholder="Enter your complete address" />
        {errors.address && <small className="error">{errors.address}</small>}
      </div>

      <div className="two-column">
        <div className="form-group">
          <label htmlFor="pincode">Pincode *</label>
          <input id="pincode" name="pincode" maxLength="6" inputMode="numeric"
value={form.pincode} onChange={handleChange} placeholder="6-digit pincode" />
          {errors.pincode && <small className="error">{errors.pincode}</small>}
        </div>

        <div className="form-group">
          <label htmlFor="phone">Phone *</label>
          <input id="phone" name="phone" maxLength="10" inputMode="numeric"
value={form.phone} onChange={handleChange} placeholder="10-digit phone number" />
          {errors.phone && <small className="error">{errors.phone}</small>}
        </div>
```

```
      </div>
      <div className="form-group">
        <label htmlFor="payment">Payment Method *</label>
        <select id="payment" name="payment" value={form.payment}
onChange={handleChange}>
          <option value="">Select payment method</option>
          <option value="Cash on Delivery">Cash on Delivery</option>
          <option value="UPI">UPI</option>
          <option value="Card">Card</option>
        </select>
        {errors.payment && <small className="error">{errors.payment}</small>}
        {errors.cart && <small className="error">{errors.cart}</small>}
      </div>
      <button className="primary-button submit-button" type="submit">
        Place Order
      </button>
    </form>
  </section>
 );
}
export default CheckoutForm;
```

Navbar.jsx

```
function Navbar({ cartCount, onCartClick }) {
 return (
  <header className="navbar">
    <div className="brand">🛒 Cartly</div>
```

```
      <nav>
        <a href="#home">Home</a>
        <a href="#products">Products</a>
        <button className="cart-button" onClick={onCartClick}>
          🛒 Cart <span className="badge">{cartCount}</span>
        </button>
      </nav>
    </header>
  );
}
export default Navbar;
```

Productcard.jsx

```
function ProductCard({ product, addToCart, onDetails }) {
  return (
    <article className="product-card">
      <img src={product.image} alt={product.name} />
      <div className="product-content">
        <span className="category">{product.category}</span>
        <h3>{product.name}</h3>
        <p className="price">₹{product.price.toLocaleString("en-IN")}</p>
        <div className="card-actions">
          <button className="secondary-button" onClick={() => onDetails(product)}>
            View Details
          </button>
          <button className="primary-button" onClick={() => addToCart(product)}>
            Add to Cart
```

```
        </button>
       </div>
     </div>
   </article>
  );
}
export default ProductCard;
```

Productlist.jsx

```
import products from "../products";
import ProductCard from "./ProductCard";
function ProductList({ addToCart, onDetails }) {
  return (
    <div className="product-grid">
      {products.map((product) => (
        <ProductCard
          key={product.id}
          product={product}
          addToCart={addToCart}
          onDetails={onDetails}
        />
      ))}
    </div>
  );
}

export default ProductList;
```